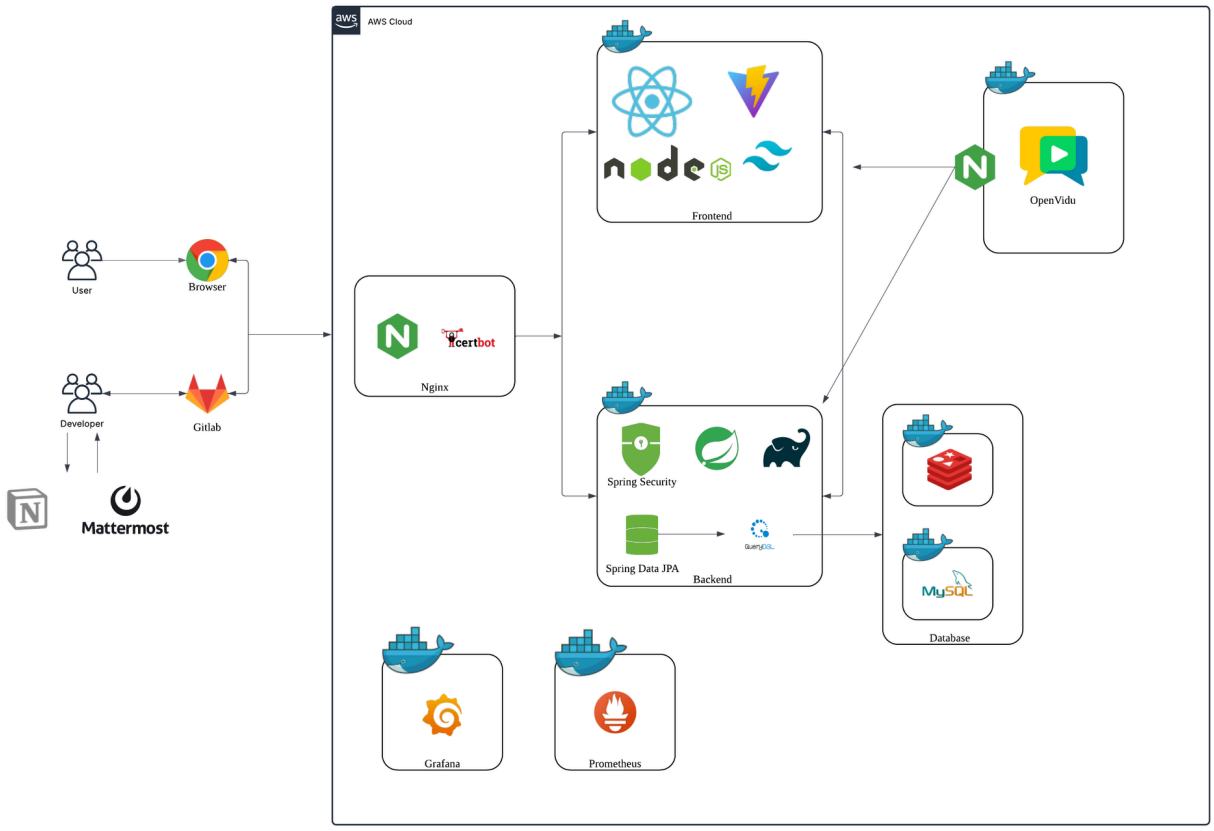
Porting Manual

시스템 아키텍처



프로젝트 사용 도구

- 이슈 관리 : JIRA

- 형상 관리 : Gitlab
- 커뮤니케이션 : Notion, Slack, MatterMost
- 디자인 : Figma

시스템 환경 정보

호스트 환경 (EC2 - Ubuntu 22.04.1)

- **JVM**: OpenJDK 17.0.16 (build 17.0.16+8-Ubuntu-0ubuntu122.04.1)
  - VM: OpenJDK 64-Bit Server VM (mixed mode, sharing)
- **웹서버**: Nginx 1.18.0 (Ubuntu)
- **컨테이너**: Docker 28.3.3 (build 980b856), Docker Compose 1.29.2

애플리케이션 서버 (Docker 컨테이너)

- **WAS**: Spring Boot (Docker 컨테이너)
  - 베이스 이미지: Amazon Corretto 24 JDK
  - 서버 포트: 8080
  - 컨텍스트 패스: /
  - 프로필: docker
  - 네트워크 모드: host
  - 헬스체크 엔드포인트: /manage/health

프론트엔드 서버 (Docker 컨테이너)

- **런타임**: Node.js 20-alpine
- **빌드 도구**: Vite
- **패키지 매니저**: npm
- **포트**: 4173 (개발용)

데이터베이스 & 캐시 (Docker 컨테이너)

- **MySQL**: 8.0
  - 문자셋: utf8mb4
  - 콜레이션: utf8mb4_unicode_ci
- **Redis**: 7-alpine
  - AOF 로깅 활성화
  - 연결 풀: Lettuce
  - 최대 활성 연결: 8개
  - 비밀번호 인증 사용

모니터링 & 메트릭

- **메트릭 시스템**: Prometheus
- **수집 주기**: 30초
- **활성화된 메트릭**: HTTP 요청, JVM 메모리, GC, 스레드, Tomcat
- **응답시간 백분위수**: 50%, 95%, 99%

빌드 도구

- **Backend**: Gradle (멀티스테이지 Docker 빌드)
- **Frontend**: npm + Vite

EC2 서버 정보

- 인스턴스 타입: `t3.medium`
- 리전: `ap-northeast-2`
- 가용영역: `ap-northeast-2a`
- 프라이빗 IP: `172.26.15.83`
- 퍼블릭 IP: `52.78.xxx.xxx`
- 도메인: `https://i13d107.p.ssafy.io/`

포트번호

웹 서비스 포트

- **80**: HTTP (Nginx)
- **443**: HTTPS (Nginx)
- **8080**: Spring Boot API 서버
- **3000, 3001**: 프론트엔드 개발 서버 (Docker)

WebRTC / 화상통화 포트 (OpenVidu)

- **3478**: STUN/TURN 서버 (TCP/UDP)
- **5442**: OpenVidu Node.js 서비스 (TCP)
- **5443**: OpenVidu Java 서비스 (TCP)
- **8888**: Kurento Media Server (TCP)
- **40000-57000** : Web RTC 미디어 스트리밍 서비스
- **57001-65535** : 확장 미디어 포트 범위

데이터베이스 & 캐시 (내부 전용)

- **3306**: MySQL (127.0.0.1)
- **6379**: Redis (127.0.0.1)

모니터링 & 관리

- **9090**: Prometheus 메트릭 수집
- **9080, 9443**: 추가 Nginx 서비스

기타 서비스

- **22**: SSH
- **53**: DNS (systemd-resolve, 127.0.0.53)
- **8989**: Apache HTTP 서버
- **8988**: Java 애플리케이션
- **29418**: Git 서비스 (Gerrit)
- **323**: NTP (chronyd)
- **68**: DHCP 클라이언트

```
ubuntu@ip-172-26-1-124:~/S13P11D107$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22                         ALLOW IN    Anywhere
80                         ALLOW IN    Anywhere
44                         ALLOW IN    Anywhere
8989                       ALLOW IN    Anywhere
443                        ALLOW IN    Anywhere
8090                       ALLOW IN    Anywhere
3478/tcp                   ALLOW IN    Anywhere
3478/udp                   ALLOW IN    Anywhere
40000:57000/tcp            ALLOW IN    Anywhere
40000:57000/udp            ALLOW IN    Anywhere
57001:65535/tcp            ALLOW IN    Anywhere
57001:65535/udp            ALLOW IN    Anywhere
9080                       ALLOW IN    Anywhere
9443                       ALLOW IN    Anywhere
9090                       ALLOW IN    Anywhere
3001                       ALLOW IN    Anywhere
8080                       ALLOW IN    Anywhere
22 (v6)                    ALLOW IN    Anywhere (v6)
80 (v6)                    ALLOW IN    Anywhere (v6)
44 (v6)                    ALLOW IN    Anywhere (v6)
8989 (v6)                  ALLOW IN    Anywhere (v6)
443 (v6)                   ALLOW IN    Anywhere (v6)
8090 (v6)                  ALLOW IN    Anywhere (v6)
3478/tcp (v6)              ALLOW IN    Anywhere (v6)
3478/udp (v6)              ALLOW IN    Anywhere (v6)
40000:57000/tcp (v6)       ALLOW IN    Anywhere (v6)
40000:57000/udp (v6)       ALLOW IN    Anywhere (v6)
57001:65535/tcp (v6)       ALLOW IN    Anywhere (v6)
57001:65535/udp (v6)       ALLOW IN    Anywhere (v6)
9080 (v6)                  ALLOW IN    Anywhere (v6)
```

배포 방법

원격접속 방법

```
ssh -i {YOUR_PEM_KEY} ubuntu@i13d107.p.ssafy.io
```

패키지 관리자 업데이트

```
sudo apt-get update
sudo apt-get upgrade
```

---

dokcer 다운로드

1. Set up Docker's `apt` repository

```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
```

2. install the Docker packages.

```
 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

3. Verify that the installation is successful by running the `hello-world` image:

```
sudo docker run hello-world
```

---

certbot 다운로드

```
sudo apt update
sudo apt install certbot python3-certbot-nginx
```

OpenVidu 다운로드

openvidu 다운로드 공식문서 : https://docs.openvidu.io/en/stable/deployment/ce/on-premises/

⭐⭐⭐⭐ nginx를 먼저 다운로드를 하면 포트 충돌이 납니다!

Open these ports (in section Close ports to avoid external attacks you have an UFW sample to configure a firewall)

- **22 TCP**: to connect using SSH to admin OpenVidu.
- **80 TCP**: if you select Let's Encrypt to generate an SSL certificate this port is used by the generation process.
- **443 TCP**: OpenVidu server and application are published by default in standard https port.
- **3478 TCP+UDP**: used by STUN/TURN server to resolve clients IPs.
- **40000 - 57000 TCP+UDP**: used by Kurento Media Server to establish media connections.
- **57001 - 65535 TCP+UDP**: used by TURN server to establish relayed media connections.

```
sudo su

cd /opt

curl https://s3-eu-west-1.amazonaws.com/aws.openvidu.io/install_openvidu_latest.sh | bash

=========================================
Openvidu Platform successfully installed.
=========================================

1. Go to openvidu folder:
$ cd openvidu
```

```
2. Configure DOMAIN_OR_PUBLIC_IP and OPENVIDU_SECRET in .env file:
$ nano .env

3. Start OpenVidu
$ ./openvidu start

For more information, check:
https://docs.openvidu.io/en/stable/deployment/ce/on-premises/
```

.env파일

```
# OpenVidu configuration
# ----------------------
# Documentation: https://docs.openvidu.io/en/stable/reference-docs/openvid
u-config/

# NOTE: This file doesn't need to quote assignment values, like most shells d
o.
# All values are stored as-is, even if they contain spaces, so don't quote them.

# Domain name. If you do not have one, the public IP of the machine.
# For example: 198.51.100.1, or openvidu.example.com
DOMAIN_OR_PUBLIC_IP=i13d107.p.ssafy.io

# OpenVidu SECRET used for apps to connect to OpenVidu server and users t
o access to OpenVidu Dashboard
OPENVIDU_SECRET=MY_SECRET

# Certificate type:
# - selfsigned:  Self signed certificate. Not recommended for production use.
#                Users will see an ERROR when connected to web page.
# - owncert:     Valid certificate purchased in a Internet services company.
#                Please put the certificates files inside folder ./owncert
#                with names certificate.key and certificate.cert
```

```
# - letsencrypt: Generate a new certificate using letsencrypt. Please set the
#                required contact email for Let's Encrypt in LETSENCRYPT_EMAIL
#                variable.
CERTIFICATE_TYPE=letsencrypt

# If CERTIFICATE_TYPE=letsencrypt, you need to configure a valid email for n
otifications
LETSENCRYPT_EMAIL=pr0104@naver.com

# Proxy configuration
# If you want to change the ports on which openvidu listens, uncomment the f
ollowing lines

# Allows any request to http://DOMAIN_OR_PUBLIC_IP:HTTP_PORT/ to be aut
omatically
# redirected to https://DOMAIN_OR_PUBLIC_IP:HTTPS_PORT/.
# WARNING: the default port 80 cannot be changed during the first boot
# if you have chosen to deploy with the option CERTIFICATE_TYPE=letsencry
pt
# HTTP_PORT=80

# Changes the port of all services exposed by OpenVidu.
# SDKs, REST clients and browsers will have to connect to this port
# HTTPS_PORT=443

# Old paths are considered now deprecated, but still supported by default.
# OpenVidu Server will log a WARN message every time a deprecated path is
called, indicating
# the new path that should be used instead. You can set property SUPPORT_D
EPRECATED_API=false
# to stop allowing the use of old paths.
# Default value is true
# SUPPORT_DEPRECATED_API=false

# If true request to with www will be redirected to non-www requests
# Default value is false
```

```
# REDIRECT_WWW=false

# How many workers to configure in nginx proxy.
# The more workers, the more requests will be handled
# Default value is 10240
# WORKER_CONNECTIONS=10240

# Access restrictions
# In this section you will be able to restrict the IPs from which you can access
to
# Openvidu API and the Administration Panel
# WARNING! If you touch this configuration you can lose access to the platfor
m from some IPs.
# Use it carefully.

# This section limits access to the /dashboard (OpenVidu CE) and /inspector
(OpenVidu Pro) pages.
# The form for a single IP or an IP range is:
# ALLOWED_ACCESS_TO_DASHBOARD=198.51.100.1 and ALLOWED_ACCESS_
TO_DASHBOARD=198.51.100.0/24
# To limit multiple IPs or IP ranges, separate by commas like this:
# ALLOWED_ACCESS_TO_DASHBOARD=198.51.100.1, 198.51.100.0/24
# ALLOWED_ACCESS_TO_DASHBOARD=

# This section limits access to the Openvidu REST API.
# The form for a single IP or an IP range is:
# ALLOWED_ACCESS_TO_RESTAPI=198.51.100.1 and ALLOWED_ACCESS_TO_R
ESTAPI=198.51.100.0/24
# To limit multiple IPs or or IP ranges, separate by commas like this:
# ALLOWED_ACCESS_TO_RESTAPI=198.51.100.1, 198.51.100.0/24
# ALLOWED_ACCESS_TO_RESTAPI=

# Whether to enable recording module or not
OPENVIDU_RECORDING=false

# Use recording module with debug mode.
```

```
OPENVIDU_RECORDING_DEBUG=false

# Openvidu Folder Record used for save the openvidu recording videos. Chan
ge it
# with the folder you want to use from your host.
OPENVIDU_RECORDING_PATH=/opt/openvidu/recordings

# System path where OpenVidu Server should look for custom recording layo
uts
OPENVIDU_RECORDING_CUSTOM_LAYOUT=/opt/openvidu/custom-layout

# if true any client can connect to
# https://OPENVIDU_SERVER_IP:OPENVIDU_PORT/recordings/any_session_fil
e.mp4
# and access any recorded video file. If false this path will be secured with
# OPENVIDU_SECRET param just as OpenVidu Server dashboard at
# https://OPENVIDU_SERVER_IP:OPENVIDU_PORT
# Values: true | false
OPENVIDU_RECORDING_PUBLIC_ACCESS=false

# Which users should receive the recording events in the client side
# (recordingStarted, recordingStopped). Can be all (every user connected to
# the session), publisher_moderator (users with role 'PUBLISHER' or
# 'MODERATOR'), moderator (only users with role 'MODERATOR') or none
# (no user will receive these events)
OPENVIDU_RECORDING_NOTIFICATION=publisher_moderator

# Timeout in seconds for recordings to automatically stop (and the session inv
olved to be closed)
# when conditions are met: a session recording is started but no user is publis
hing to it or a session
# is being recorded and last user disconnects. If a user publishes within the ti
meout in either case,
# the automatic stop of the recording is cancelled
# 0 means no timeout
OPENVIDU_RECORDING_AUTOSTOP_TIMEOUT=120
```

```
# Maximum video bandwidth sent from clients to OpenVidu Server, in kbps.
# 0 means unconstrained
OPENVIDU_STREAMS_VIDEO_MAX_RECV_BANDWIDTH=1000

# Minimum video bandwidth sent from clients to OpenVidu Server, in kbps.
# 0 means unconstrained
OPENVIDU_STREAMS_VIDEO_MIN_RECV_BANDWIDTH=300

# Maximum video bandwidth sent from OpenVidu Server to clients, in kbps.
# 0 means unconstrained
OPENVIDU_STREAMS_VIDEO_MAX_SEND_BANDWIDTH=1000

# Minimum video bandwidth sent from OpenVidu Server to clients, in kbps.
# 0 means unconstrained
OPENVIDU_STREAMS_VIDEO_MIN_SEND_BANDWIDTH=300

# All sessions of OpenVidu will try to force this codec. If OPENVIDU_STREAM
S_ALLOW_TRANSCODING=true
# when a codec can not be forced, transcoding will be allowed
# Values: MEDIA_SERVER_PREFERRED, NONE, VP8, VP9, H264
# Default value is MEDIA_SERVER_PREFERRED
# OPENVIDU_STREAMS_FORCED_VIDEO_CODEC=MEDIA_SERVER_PREFERRE
D

# Allow transcoding if codec specified in OPENVIDU_STREAMS_FORCED_VID
EO_CODEC can not be applied
# Values: true | false
# Default value is false
# OPENVIDU_STREAMS_ALLOW_TRANSCODING=false

# true to enable OpenVidu Webhook service. false' otherwise
# Values: true | false
OPENVIDU_WEBHOOK=false

# HTTP endpoint where OpenVidu Server will send Webhook HTTP POST mes
```

```
sages
# Must be a valid URL: http(s)://ENDPOINT
#OPENVIDU_WEBHOOK_ENDPOINT=

# List of headers that OpenVidu Webhook service will attach to HTTP POST m
essages
#OPENVIDU_WEBHOOK_HEADERS=

# List of events that will be sent by OpenVidu Webhook service
# Default value is all available events
OPENVIDU_WEBHOOK_EVENTS=[sessionCreated,sessionDestroyed,participa
ntJoined,participantLeft,webrtcConnectionCreated,webrtcConnectionDestroy
ed,recordingStatusChanged,filterEventDispatched,mediaNodeStatusChanged,
nodeCrashed,nodeRecovered,broadcastStarted,broadcastStopped]

# How often the garbage collector of non active sessions runs.
# This helps cleaning up sessions that have been initialized through
# REST API (and maybe tokens have been created for them) but have had no
users connected.
# Default to 900s (15 mins). 0 to disable non active sessions garbage collector
OPENVIDU_SESSIONS_GARBAGE_INTERVAL=900

# Minimum time in seconds that a non active session must have been in existe
nce
# for the garbage collector of non active sessions to remove it. Default to 360
0s (1 hour).
# If non active sessions garbage collector is disabled
# (property 'OPENVIDU_SESSIONS_GARBAGE_INTERVAL' to 0) this property is
ignored
OPENVIDU_SESSIONS_GARBAGE_THRESHOLD=3600

# Call Detail Record enabled
# Whether to enable Call Detail Record or not
# Values: true | false
OPENVIDU_CDR=false
```

```
# Path where the cdr log files are hosted
OPENVIDU_CDR_PATH=/opt/openvidu/cdr

# Kurento Media Server image
# --------------------------
# Docker hub kurento media server: https://hub.docker.com/r/kurento/kurento
-media-server
# Uncomment the next line and define this variable with KMS image that you
want use
# KMS_IMAGE=kurento/kurento-media-server:7.1.1

# Kurento Media Server Level logs
# -------------------------------
# Uncomment the next line and define this variable to change
# the verbosity level of the logs of KMS
# Documentation: https://doc-kurento.readthedocs.io/en/stable/features/loggi
ng.html
# KMS_DOCKER_ENV_GST_DEBUG=

# Openvidu Server Level logs
# --------------------------
# Uncomment the next line and define this variable to change
# the verbosity level of the logs of Openvidu Service
# RECOMENDED VALUES: INFO for normal logs DEBUG for more verbose logs
# OV_CE_DEBUG_LEVEL=INFO

# Java Options
# ------------------------
# Uncomment the next line and define this to add
# options to java command
# Documentation: https://docs.oracle.com/cd/E37116_01/install.111210/e23737/
configuring_jvm.htm#OUDIG00058
# JAVA_OPTIONS=-Xms2048m -Xmx4096m -Duser.timezone=UTC
```

```
HTTP_PORT=9080
HTTPS_PORT=9443
```

openvidu 실행

```
./openvidu start

## 실행화면
eabaf7c34db0   kurento/kurento-media-server:7.1.1   "/entrypoint.sh"        7 da
ys ago    Up 7 days (healthy)
eabaf7c34db0_openvidu-kms-1
f65ae95e5c1e   openvidu/openvidu-proxy:2.31.0       "/docker-entrypoint...."
7 days ago    Up 7 days
openvidu-nginx-1
761c5a493b37   openvidu/openvidu-call:2.31.0        "docker-entrypoint.s..."   7
days ago    Up 7 days
openvidu-app-1
1beaa3ce2582   openvidu/openvidu-coturn:2.31.0       "docker-entrypoint.s..."
7 days ago    Up 7 days             0.0.0.0:3478→3478/tcp, 0.0.0.0:3478→3478/
udp, [::]:3478→3478/tcp, [::]:3478→3478/udp, 5349/tcp, 5349/udp   1beaa3c
e2582_openvidu-coturn-1
7f1442655556   openvidu/openvidu-server:2.31.0       "/usr/local/bin/entr..."   7
days ago    Up 7 days
openvidu-openvidu-server-1
```

---

nginx

nginx 설치방법

```
# 패키지 목록 업데이트
sudo apt update
# Nginx 설치
sudo apt install nginx
```

```
# Nginx 문법 확인
sudo nginx -t
# Nginx 서비스 시작
sudo systemctl start nginx
# 부팅시 자동 시작 설정
sudo systemctl enable nginx
# 상태 확인
sudo systemctl status nginx
```

nginx 프록시 설정하기

```
sudo nano /etc/nginx/sites-available/default
```

etc/nginx/sites-available/default

```
# HTTP → HTTPS 리다이렉트
server {
    listen 80; # IPv4
    listen [::]:80; # IPv6
    server_name i13d107.p.ssafy.io;
    return 301 https://$host$request_uri;
}

# HTTPS 메인 서버
server {
    listen 443 ssl http2;
    listen [::]:443 ssl http2;
    server_name i13d107.p.ssafy.io;

    # SSL 설정 (Certbot이 관리)
    ssl_certificate /etc/letsencrypt/live/i13d107.p.ssafy.io/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/i13d107.p.ssafy.io/privkey.pem;
```

```
    include /etc/letsencrypt/options-ssl-nginx.conf;
    ssl_dhparam /etc/letsencrypt/ssl-dhparams.pem;

    # 추가 보안 헤더
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header Strict-Transport-Security "max-age=31536000; includeSubDo
mains" always;

    # 파일 업로드 크기 제한
    client_max_body_size 100M;

    # Gzip 압축
    gzip on;
    gzip_vary on;
    gzip_min_length 1024;
    gzip_types text/plain text/css text/xml text/javascript application/javascript
application/xml+rss application/json;

    # API 라우팅 (Spring Boot 백엔드) - 포트 8080으로 수정
    location /api/ {
        proxy_pass http://localhost:8080;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache_bypass $http_upgrade;

        # JWT 및 CORS 관련 헤더 유지
        proxy_set_header Authorization $http_authorization;
        proxy_pass_header Authorization;
    }
```

```
# Spring Actuator (내부 관리용) - 포트 8080으로 수정
location /manage/ {
    proxy_pass http://localhost:8080;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;

    # 관리용 엔드포인트 - 필요시 IP 제한 가능
    allow 10.0.0.0/8;      # Docker 내부 네트워크
    allow 172.16.0.0/12;   # Docker 기본 네트워크
    deny all;
}

# OpenVidu WebSocket 및 API - HTTPS 포트 9443으로 수정
location /openvidu/ {
    proxy_pass https://localhost:9443;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_cache_bypass $http_upgrade;

    # SSL 검증 비활성화 (OpenVidu 자체 서명 인증서)
    proxy_ssl_verify off;
    proxy_ssl_session_reuse off;

    # 타임아웃 설정 (화상통화용 - 더 긴 시간 설정)
    proxy_connect_timeout 300s;
    proxy_send_timeout 300s;
    proxy_read_timeout 300s;
```

```
    # 대용량 파일 업로드 지원 (영상 파일)
    client_max_body_size 500M;
}

# WebSocket 지원 (Spring WebSocket) - 포트 8080으로 수정
location /ws/ {
    proxy_pass http://localhost:8080;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "upgrade";
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_cache_bypass $http_upgrade;
}

# Frontend React 앱 (메인 경로) - 포트 3000으로 유지
location / {
    proxy_pass http://localhost:3000;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;

    # SPA 지원을 위한 설정
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'upgrade';
    proxy_cache_bypass $http_upgrade;

    # SPA 라우팅을 위한 fallback
    proxy_intercept_errors on;
    error_page 404 = @fallback;
}
```

```
    # SPA fallback (History API 지원)
    location @fallback {
        proxy_pass http://localhost:3000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    # 로그 설정
    access_log /var/log/nginx/aiders_access.log;
    error_log /var/log/nginx/aiders_error.log;
}
```

.env

```
# Port Configuration
REDIS_PORT=6379
MYSQL_PORT=3306
FRONTEND_HOST_PORT=3000
FRONTEND_CONTAINER_PORT=5173

# Database Configuration
MYSQL_ROOT_PASSWORD=SJ0pbTsi2024
MYSQL_DATABASE=mydb
DB_USERNAME=root

# Redis Configuration
REDIS_PASSWORD=SJ0pbTsi2024

# OpenVidu Configuration
```

```
OPENVIDU_SECRET=MY_SECRET
OPENVIDU_URL=http://localhost:5443

# Spring Configuration
SPRING_PROFILES_ACTIVE=docker

# Application URLs
VITE_API_BASE_URL=https://i13d107.p.ssafy.io
VITE_WS_BASE_URL=wss://i13d107.p.ssafy.io/ws

# API Keys
KAKAO_MAP_API_KEY={YOUR_KAKAO_MAP_KEY}
```

aiders-frontend/.env

```
VITE_KAKAO_MAP_API_KEY ={YOUR_KAKAO_MAP_KEY}
VITE_WS_BASE_URL="wss://i13d107.p.ssafy.io/ws"
```

application-docker.properties

```
spring.application.name=aiders

#it will be set build date by gradle. if this value is @build.date@, front-end is d
evelopment mode
build.date=@build.date@
server.port=8080
server.address=0.0.0.0
server.servlet.contextPath=/
# Charset of HTTP requests and responses. Added to the "Content-Type" hea
der if not set explicitly.
server.servlet.encoding.charset=UTF-8
# Enable http encoding support.
server.servlet.encoding.enabled=true
```

```
# Force the encoding to the configured charset on HTTP requests and respon
ses.
server.servlet.encoding.force=true

# for SPA
spring.web.resources.static-locations=classpath:/dist/
spa.default-file=/dist/index.html
spring.web.resources.add-mappings=false

#database - will be overridden by environment variables
spring.jpa.hibernate.naming.implicit-strategy=org.springframework.boot.orm.j
pa.hibernate.SpringImplicitNamingStrategy
spring.jpa.hibernate.ddl-auto=update
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL8Dialect
spring.jpa.properties.hibernate.format_sql=true
spring.jpa.properties.hibernate.default_batch_fetch_size=1000
spring.jpa.properties.hibernate.connection.characterEncoding=UTF-8
spring.jpa.properties.hibernate.connection.useUnicode=true
spring.jpa.properties.hibernate.connection.charSet=UTF-8
spring.data.web.pageable.one-indexed-parameters=true
spring.datasource.url=jdbc:mysql://localhost:3306/mydb?useUnicode=true&c
haracterEncoding=UTF-8&serverTimezone=Asia/Seoul&useSSL=false&allowP
ublicKeyRetrieval=true
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.hikari.username=root
spring.datasource.hikari.password=SJ0pb#Tsi

# jwt
spring.jwt.secret=Qp$N,SPFgnrb+<taJgdtuj<>&`Qg&]Sz-TG[c%[>;W{I?yP#W,
fXw%6,7j}:llb

# unit is ms. 15 * 24 * 60 * 60 * 1000 = 15days
spring.jwt.expiration=1296000000

spring.devtools.livereload.enabled=false
```

```
#gzip compression
server.compression.enabled=true
server.compression.mime-types=application/json,application/xml,text/html,te
xt/xml,text/plain,application/javascript,text/css

#for health check
management.endpoints.web.base-path=/manage
management.health.db.enabled=true
management.health.default.enabled=true
management.health.diskspace.enabled=true
management.endpoint.health.show-details=always
management.endpoint.health.show-components=always
management.endpoints.web.exposure.include=*

# Prometheus metric
management.endpoint.prometheus.enabled=true
management.metrics.export.prometheus.enabled=true

# metric 주기 설정
management.metrics.export.prometheus.step=30s

# Response Time 메트릭 활성화
management.metrics.web.server.request.autotime.enabled=true
management.metrics.web.server.request.autotime.percentiles=0.5,0.95,0.99
management.metrics.web.server.request.autotime.percentiles-histogram=true

# Thread 메트릭 세부화
management.metrics.jvm.threads.enabled=true

# Tomcat 메트릭 활성화
management.metrics.tomcat.enabled=true

# HTTP 요청 메트릭 세부 설정
management.metrics.web.server.request.metric-name=http.server.requests
management.metrics.web.server.request.autotime.max-expected-time=10s
```

```
# 메모리 메트릭 활성화
management.metrics.jvm.memory.enabled=true
management.metrics.jvm.gc.enabled=true

#for openVidu - will be overridden by environment variables
openvidu.url=http://aiders-openvidu:4443/
openvidu.secret=MY_SECRET

# Security Debug Logging
logging.level.org.springframework.security=INFO
logging.level.org.springframework.web=INFO
logging.level.team1234.aiders=INFO
logging.level.org.springframework.security.web.csrf=INFO

# Redis - Docker container hostname
spring.data.redis.host=127.0.0.1
spring.data.redis.password=SJ0pbTsi2024
spring.data.redis.port=6379
spring.data.redis.timeout=2000ms
spring.data.redis.lettuce.pool.max-active=8
spring.data.redis.lettuce.pool.max-wait=-1ms
spring.data.redis.lettuce.pool.max-idle=8
spring.data.redis.lettuce.pool.min-idle=0


#OpenAI
spring.ai.openai.base-url=https://gms.ssafy.io/gmsapi/api.openai.com
spring.ai.openai.api-key=S13P12D107-3d8f3400-ed6e-4e4a-9f79-efe96b031
6ca
spring.ai.openai.chat.options.model=gpt-4.1
spring.ai.openai.chat.options.temperature=0.2

# TimeZone
spring.jackson.time-zone=Asia/Seoul
spring.jackson.date-format=yyyy-MM-dd HH:mm:ss
```

docker-compose로 container 실행하기

```
cd S13P11D107
```

```
docker-compose up --build -d
```

```
docker ps

CONTAINER ID   IMAGE                          COMMAND                  CREATED
STATUS              PORTS                                                                                   
NAMES
b872c987ac4a   s13p11d107_aiders-frontend          "docker-entrypoint.s..."   5
hours ago   Up 5 hours            4173/tcp, 0.0.0.0:3000->5173/tcp, [::]:3000->51
73/tcp                                                 aiders-frontend
7199c11da92c   s13p11d107_aiders-app               "java -jar app.jar"      5 hours a
go   Up 5 hours (healthy)
aiders-backend
4aa9f077b79f   mysql:8.0                           "docker-entrypoint.s..."   5 hours ag
o   Up 5 hours            127.0.0.1:3306->3306/tcp, 33060/tcp
aiders-mysql
db94eca914a6   prom/prometheus:latest              "/bin/prometheus --c..."   5
hours ago   Up 5 hours
aiders-prometheus
6c4c97e587d6   grafana/grafana:latest              "/run.sh"                5 hours ago
Up 5 hours            0.0.0.0:3001->3000/tcp, [::]:3001->3000/tcp
aiders-grafana
2193bbcbeb4b   redis:7-alpine                      "docker-entrypoint.s..."   5 hours
ago   Up 5 hours            127.0.0.1:6379->6379/tcp
aiders-redis
```

- aiders-app의 STATUS 옆에 (healthy)라는 값이 나와야 backend가 실행이 됨

dumpfile으로 데이터 입력 방법

hospital_final.sql, ambulance_final.sql, firestation_final.sql, emergency_bed.sql, hospital_department.sql의 파일을 scp를 이용해 ec2로 전송한다.

```
scp -i your-key.pem hospital_final.sql ambulance_final.sql firestation_final.sql
emergency_bed.sql hospital_department.sql ubuntu@i13d107.p.ssafy.io:~/
```

결과화면

```
ubuntu@ip-172-26-1-124:~$ ls
S13P11D107            firestation_final.sql
ambulance_final.sql   hospital_department.sql
emergency_bed.sql     hospital_final.sql
```

⭐⭐⭐⭐⭐⭐⭐⭐ 5개의 파일은 `hospital_final.sql → emergency_bed.sql → hospital_department.sql → firestation_final.sql → ambulance_final.sql` 의 순서로 넣어야한다.

dump file 입력

```
# 1. hospital_final.sql
docker exec -i aiders-mysql mysql -u root -p"SJ0pb#Tsi" mydb < hospital_fin
al.sql

# 2. emergency_bed.sql
docker exec -i aiders-mysql mysql -u root -p"SJ0pb#Tsi" mydb < emergency_
bed.sql

# 3. hospital_department.sql
docker exec -i aiders-mysql mysql -u root -p"SJ0pb#Tsi" mydb < hospital_de
partment.sql

# 4. firestation_final.sql
```

```
docker exec -i aiders-mysql mysql -u root -p"SJ0pb#Tsi" mydb < firestation_final.sql

# 5. ambulance_final.sql
docker exec -i aiders-mysql mysql -u root -p"SJ0pb#Tsi" mydb < ambulance_final.sql
```

로그인 계정

테스트 계정

👮 관리자: admin / admin
🚒 소방서: 6110582 / f63f9527-a468-4fcb-bf4c-836a3313ecd6
🏥 병원: A1200028 / 3e9cf29d-93de-46b6-bbb1-a97ac1fac65f
🚑 구급차: 998버4200 / 8dedb374-c0d8-4525-85ad-e48d4372bc0d ← AI 모델 자동 다운로드

관리자

- admin
- admin

소방서

- 6110582
- f63f9527-a468-4fcb-bf4c-836a3313ecd6

병원

- A1200028
- 3e9cf29d-93de-46b6-bbb1-a97ac1fac65f

구급차

- 998버4200
- 8dedb374-c0d8-4525-85ad-e48d4372bc0d